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MAT1001 Midterm Examination

Saturday, October 29, 2022

Time: 9:30 - 11:30 AM

Notes and Instructions

1. *No books, no notes, no dictionaries, and no calculators.*
2. *The total score of this examination is 100.*
3. *There are **11** questions (with parts) in total.*
4. *The symbol $[N]$ at the beginning of a question indicates that the question is worth N points.*
5. *Answer all questions on the **answer book**.*
6. *Show your intermediate steps **except Questions 1 and 2** — answers without intermediate steps will receive minimal (or even no) marks.*

MAT1001 Midterm Questions

1. [10] True or False (in general)? No explanation is required.

(i) If $f(4) = -\pi$, $f(8) = \sqrt{2}$, and f is differentiable everywhere, then the equation $f(x) = 0$ must have at least one solution on the interval $(4, 8)$. True

(ii) Suppose that $y = f(x)$ is increasing and differentiable on the real line. Then for any x , if $\Delta x > 0$ then $\Delta y > dy$. False

(iii) If $f(x) < g(x)$ for all $x \in (-1, 1)$, and the limits of f and g both exist as x approaches 0, then we have $\lim_{x \rightarrow 0} f(x) < \lim_{x \rightarrow 0} g(x)$. False

(iv) The function $y = \frac{\sin(x)}{x}$ has a horizontal asymptote as $x \rightarrow \infty$. True

(v) Suppose that $g(x) \leq f(x) \leq h(x)$ for all $x > 0$, and the limits of g and h both exist as x approaches 1. Then $\lim_{x \rightarrow 1} f(x)$ must exist. True

2. [16] Short questions: no explanation is required.

(i) Supposing a function $y = f(x)$ defined on the real line has a second derivative $f''(x)$ that is negative for $x < 0$ and is positive for $x > 0$. However, both $f'(x)$ and $f''(x)$ fail to exist at $x = 0$, and there is a vertical tangent there. Which one of the following statements **must** be correct?

- A) The point $x = 0$ gives a local maximum.
- B) The point $x = 0$ gives a local minimum.
- C) The point $(0, f(0))$ gives an inflection point.
- D) The point $x = 0$ is a discontinuity.

(ii) The function $f(x) = |x - 3|$ is differentiable over which of the following domains?

- A) $(-4, 4)$
- B) $(-5, 1)$
- C) $(-\infty, +\infty)$
- D) $(-1, 5)$

$$2x \sin^3(4x) + x^2 3 \sin^2(4x) \cos(4x) \cdot 4$$

(iii) Compute $f'(x)$, where $f(x) = x^2 \sin^3(4x)$. No need to simplify.

(iv) Suppose that $F(x) = f(x)g(x)$, where f and g have derivatives of all orders. Express F'' in terms of f , g , and their first and second derivatives.

$$F' = f(x)g'(x) + f'(x)g(x)$$

$$F'' = f''(x)g(x) + f'(x)g'(x) + f'(x)g'(x) + f(x)g''(x)$$

$$F'' = f''(x)g(x) + 2f'(x)g'(x) + f(x)g''(x)$$

(v) Assume that $u = u(x)$, $v = v(x)$ and $w = w(x)$ are three functions of x , with $w(x)$ never zero. Compute the derivative of $\frac{u(x) + v^2(x)}{w(x)}$ with respect of x in terms of u, v, w, u', v' , and w' . No need to simplify.

$$\frac{w(x)(u'(x) + 2v(x)v'(x)) + (u(x) + v^2(x))w'(x)}{(w(x))^2}$$

(vi) A particle vibrates horizontally on the x -axis. Its position at time t is given by $x(t) = 8 \sin t$, where t is in seconds and x is in centimeters.

Find the velocity and acceleration of the particle at time $t = \frac{2\pi}{3}$.

$$v(t) = 8 \cos t, \quad a(t) = -8 \sin t = -4\sqrt{3}$$

(vii) For (vi) above, in what direction is the particle moving at $t = \frac{2\pi}{3}$?

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(viii) Find the function $y = f(x)$ defined for $x > 0$ that satisfies $y' = \frac{1+x^3}{x^2}$ and $y(2) = 1$.

$$y' = \frac{1}{x^2} + x$$

$$y = -\frac{1}{x} + \frac{x^2}{2} - \frac{1}{2}$$

3. [4+4] Consider the equation $x^3 + 3x + 1 = 0$.

(i) Prove that the equation has a solution $x = c$ with $|c| < 1$.

$f(1) = 9, f(-1) = -3 \rightarrow$ IVT, There exists $c \in (-1, 1)$ s.t. $f(c) = 0$

(ii) Consider using Newton's method to estimate the solution. Starting with $x_0 = 0$, find x_1 and x_2 . Express your answers in the form n/m where n and m are integers.

$$x_1 = -\frac{1}{3}, x_2 = -\frac{1}{12}$$

4. [4+4+4] Evaluate the following limits. Use only methods and theories from Chapters 2, 3, or 4 in the textbook.

$$(i) \lim_{x \rightarrow 0} \frac{8x}{3 \sin(x) - x} = \lim_{x \rightarrow 0} \frac{\frac{8}{x}}{\frac{3 \sin x}{x} - 1} = \frac{8}{2} = 4$$

$$(ii) \lim_{x \rightarrow -\infty} \sqrt{x^2 + 3x} - \sqrt{x^2 - 2x}$$

$$(iii) \lim_{x \rightarrow 9} \frac{\sin(\sqrt{x} - 3)}{x - 9} = \lim_{x \rightarrow 9} \frac{\sin(\sqrt{x} - 3)}{\sqrt{x} - 3} \cdot \frac{1}{\sqrt{x} + 3} = 1 \cdot \frac{1}{6}$$

$$= \lim_{x \rightarrow 9} \frac{5x}{\sqrt{x^2 + 3x} + \sqrt{x^2 - 2x}} = \lim_{x \rightarrow 9} \frac{5x}{x \sqrt{1 + \frac{3}{x}} + x \sqrt{1 - \frac{2}{x}}} = \frac{5}{2}$$

5. [4+4] The biomass $B(t)$ of a fish population is the total mass of the members of the population at time t . Let $N(t)$ denote the number of indi-

5. [4+4] The biomass $B(t)$ of a fish population is the total mass of the members of the population at time t . Let $N(t)$ denote the number of individuals in the population at time t , and let $M(t)$ denote the average mass of a fish at time t .

$$B(t) = N(t) \cdot M(t) \rightarrow B'(t) = N(t)M'(t) + N'(t)M(t)$$

- (i) Express $B(t)$ and $B'(t)$ using the information above.
- (ii) In the case of the *black moor goldfish*, breeding occurs continually. Suppose that at time $t = 4$ weeks the population is 820 fish and is growing at a rate of 50 fish per week, while the average mass is 1.2 g and is increasing at a rate of 0.1 g/week. At what rate is the biomass increasing when $t = 4$?

$$B'(4) = 820 \cdot 0.1 + 50 \cdot 1.2 = 142 \text{ g/week}$$

6. [6] Suppose that there is a conical tank standing pointing up. The radius of the base is $R = 6$ m and the height is $H = 4$ m. Let the water run in the tank from the top at the rate of $0.5 \text{ m}^3/\text{min}$. How fast is the water level raising when the water level is half the height of the tank (2 m)?

7. [6] Suppose that a particle is travelling on the orbit $\frac{x^2}{4} + \frac{y^2}{9} = 1$. If the velocity of the particle in the x -direction is $\frac{dx}{dt} = 6$ (m/s) when $x = 1$ m and $y > 0$, find the velocity of the particle in the y -direction (which is $\frac{dy}{dt}$).

8. [4+3] Suppose that f is differentiable everywhere, with $f(-6) = 4$ and $f'(x) \leq 3$ for all x .

- (i) What is the largest possible value for $f(0)$? Prove your claim using a theorem in Chapter 4.

- (ii) Find an example of such f that gives the largest value of $f(0)$.

9. [4+3+3] Let $f(x) = \cos^2(x) - 2\sin(x)$ with domain $D = [0, 2\pi]$.

- (i) Find the closed interval(s) where $f(x)$ is strictly increasing, then find those where $f(x)$ is strictly decreasing.

- (ii) Find all the critical points of $f(x)$ in the interior of D .

- (iii) Identify the x -positions of all the local maxima and local minima of $f(x)$ in D (for each x , state whether it gives a local min or a local max).

$$\text{Min: } x = \frac{\pi}{2}, 2\pi$$

$$\text{Max: } x = \frac{3\pi}{2}, 0$$

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10. [4+2+6] Consider the function $f(x) = x^3 - 6x^2 + 6x$ defined on $[0, \infty)$.

- Find all interval(s) where f is concave up, then find those where f is concave down. (You can include the endpoints for your intervals.)
- Find all inflection points. (State their x -coordinates.)
- Sketch the graph $y = f(x)$. Indicate the following features on the graph: concavity, the x -positions of all the local and global (absolute) extrema, and the x -intercepts (zero crossings). For each extremum, indicate its type (i.e., local or global, max or min.)

11. [5] Suppose that $f(x)$ is continuous for $x \in [-1, 1]$ and $f(0) = 3$. Determine the value of the following limit, or explain why it does not exist:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt[3]{1 + f(x) \sin(x)} - 1}{x} &= \lim_{x \rightarrow 0} \frac{f(x) \sin(x)}{x \left(\sqrt[3]{(1 + f(x) \sin(x))^3} - \sqrt[3]{1 + f(x) \sin(x)} + 1 \right)} \\ &= \lim_{x \rightarrow 0} \frac{\sin(x)}{x} \cdot \frac{f(x)}{\sqrt[3]{(1 + f(x) \sin(x))^3} - \sqrt[3]{1 + f(x) \sin(x)} + 1} \\ &= 1 \cdot \frac{3}{\sqrt[3]{(1+0)^3} - \sqrt[3]{1+0} + 1} = 1 \cdot \frac{3}{3} = 1 \end{aligned}$$

MAT1001 Midterm Examination

Saturday, October 28, 2023

Time: 9:30 - 11:30 AM

Notes and Instructions

1. No books, no notes, no dictionaries, and no calculators.
2. The maximum possible score of this examination is **110**.
3. There are **13** questions (with parts), which are worth 120 points in total. **This means that you do not have to answer all the questions in order to get the full score.**
4. The symbol **[N]** at the beginning of a question indicates that the question is worth N points.
5. Write down your solutions on the **answer book**.
6. Show your intermediate steps **except Questions 1, 2, and 3** — answers without intermediate steps will receive minimal (or even no) marks.



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1. [10] True or False? No explanation is required.

(i) If $\lim_{x \rightarrow 0} |f(x)| = 0$, then $\lim_{x \rightarrow 0} f(x) = 0$. **True**

(ii) If $y = (f(x))^2$ is continuous on the real line, then $y = f(x)$ is also continuous on the real line. **False**

(iii) The graph of

$$U(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

has a vertical tangent at the point $(0, 1)$. **False**

(iv) Suppose that $y = f(x)$ is decreasing and concave up on the real line. Then for any x , if $\Delta x = dx > 0$, then $|\Delta y| < |dy|$. **True**

(v) If f is continuous on (a, b) then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in (a, b) . **False**

2. [9] For each part of this question, there is only one correct answer. Choose the correct answer. No explanation is required.

(i) Consider the function $y = f(x)$ defined over the interval $[0, 1]$ as follows:

$$f(x) = \begin{cases} x - x^2, & 0 < x < 1 \\ 1, & x = 0 \text{ and } x = 1 \end{cases}$$

Which of the following is correct?

- A) The function has three absolute maxima.
 B) The function has no local minimum and three local maxima.
 C) The function has one local minimum and two absolute maxima.
 D) The function has two local minima and one local maximum.



- (ii) Let f be twice-differentiable on $I = (a, b)$ and continuous on $[a, b]$, and let $f'' > 0$ on I . Which of the following must be correct?

Concave up

- A) For any $c \in I$, the tangent line to $y = f(x)$ at c lies below the graph of $y = f(x)$.
 B) For any $c \in I$, the tangent line to $y = f(x)$ at c lies above the graph of $y = f(x)$.
 C) The graph of f lies above the secant line joining $(a, f(a))$ and $(b, f(b))$.
 D) None of the above (A), (B), (C) is true.

- (iii) Consider using Newton's method to solve the equation $f(x) = 0$ where

$$f(x) = a - (x - b)^2.$$

Here, $a > 0$ and b is arbitrary. What initial guess x_0 below will always approximate the largest root?

- A) Choose $x_0 < a$.
 B) Choose $x_0 \leq b$.
 C) Choose $x_0 > b$.
 D) Any initial guess x_0 can guarantee the convergence.

3. [15] Short questions: no explanation is required.

$$y' = \cos(\cos(2t - \frac{\pi}{6})) \cdot (-\sin(2t - \frac{\pi}{6})) \cdot 2 \therefore y'(\frac{\pi}{6}) = 1 \cdot (-1) \cdot 2 = -2$$

- (i) Calculate the derivative of $y = \sin\left(\cos\left(2t - \frac{\pi}{6}\right)\right)$ at $t = \frac{\pi}{6}$.

- (ii) Given that $y = x^2 + 7x - 5$ and $\frac{dx}{dt} = \frac{1}{3}$ when $x = 1$, find $\frac{dy}{dt}$ at $x = 1$.

- (iii) Find the linearization of the function $f(x) = \frac{1}{1-x}$ centered at $x = 0$.
 $\frac{dy}{dt} = 2x \frac{dx}{dt} + 7 \frac{dx}{dt} = 2 \cdot 1 \cdot \frac{1}{3} + 7 \cdot \frac{1}{3} = 3$
 $L(x) = \frac{1}{1-0} + \frac{-1}{(1-0)^2} \cdot (-1) \cdot (x-0) = 1+x$

- (iv) We wish to estimate the solution to the equation $x^3 - x - 5 = 0$ using Newton's method. Supposing we take the initial estimate $x_0 = 0$, find x_1 .

$$x_1 = 0 - \frac{-5}{-1} = -5$$



$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^2 \sin(\frac{1}{h})}{h} = \lim_{h \rightarrow 0} h \sin(\frac{1}{h}) = 0$$

7. [8] Is the derivative of $f(x) = x^2 \cos(\frac{1}{x}) + 2x \sin(\frac{1}{x})$ continuous at $x=0$? DNE = 0

$$f'(x) = 2x \cos(\frac{1}{x}) - x^2 \sin(\frac{1}{x}) \cdot (-\frac{1}{x^2}) + 2 \sin(\frac{1}{x}) + 2x \cos(\frac{1}{x}) \cdot (-\frac{1}{x^2})$$

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases} \rightarrow \text{Not continuous}$$

continuous at $x=0$? Is the derivative of $g(x) = xf(x)$ continuous at $x=0$?
 Give reasons for your answers.

8. [5+4] Given the curve defined by the equation $x^2(2-y) = y^3$:

(i) Find the equation of the tangent line to the curve at $(1, 1)$.

(ii) Find $\frac{d^2y}{dx^2}$ at $(1, 1)$.

$$2x^2 - x^2y - y^3$$

$$4x - x^2 \frac{dy}{dx} - 2xy = 3y^2 \frac{dy}{dx}$$

$$\frac{dy}{dx} (4 - 2xy) = (3y^2 - 4x) \frac{dy}{dx} \rightarrow \frac{dy}{dx} = \frac{4x - 2xy}{3y^2 - 4x}$$

At $(1, 1)$, $\frac{dy}{dx} = \frac{4-2}{3-4} = \frac{1}{-1} = -1$
 Tangent line eq: $y - 1 = -1(x - 1)$

9. [6] A car braked with a constant deceleration of 16 ft/s^2 (feet per second squared), producing skid marks (刹车痕) measuring 200 ft before coming to a stop. How fast was the car traveling when the brakes were first applied? Answer this question using the theory of antidifferentiation.

10. [4+3+4] Consider the function $f(x) = x^4 - 4x^3$ defined on the real line.

(i) Determine the intervals where $f(x)$ is concave up and where $f(x)$ is concave down.

(ii) Determine the points of inflection of this function.

(iii) Determine the locations of all local maxima and minima.

11. [9] Consider the equation

$$f(x) = x^3 - 2x + c = 0$$

where c is a constant. Without solving the equation, determine the range of values of c for which:

(i) the equation has only one solution,

(ii) the equation has exactly two solutions, and

(iii) the equation has three solutions.

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of values of c for which:

- (i) the equation has only one solution,
- (ii) the equation has exactly two solutions, and
- (iii) the equation has three solutions.

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$$0 = \frac{dA}{dL_1} = \frac{1}{2}(6L_1^2 - 5LL_1 + L^2) = (2L_1 - L)(3L_1 - L) \rightarrow L_1 = \frac{L}{3}$$

$$\rightarrow \frac{1}{2}\left(\frac{L}{3} - L\right)\left(\frac{L}{3} - L\right) = \frac{1}{2}\left(1L - \frac{2}{3}L^2 + \frac{2}{3}L^2 - \frac{1}{9}L^2\right)$$

$$A = \sqrt{\frac{1}{2}\left(\frac{L}{3} - L\right)\left(\frac{L}{3} - L\right)} = \frac{L}{3}$$

$L_1, L_1, L - 2L_1$
 L

12. [7] A string of length L cm is used to form a triangle $\triangle ABC$ whose sides AB and AC are of the same length L_1 cm, where $2L_1 < L$. Find L_1 in terms of L so that the area of the triangle is maximized.

13. [2+5] Suppose the function $f(x)$ is continuous on $[0, 1]$ and twice differentiable on $(0, 1)$.

(i) Use standard linear approximation of f at $x = 0$ to approximate $f(1)$.

(ii) Show that there exist $A \in (-\frac{1}{2}, \frac{1}{2})$ and $c \in (0, 1)$ such that

$$f(1) = f(0) + f'(0)\left(\frac{1}{2}\right) + Af''(c).$$

$$\text{MVT: } \frac{f(1) - f(0)}{1 - 0} = f'(d) \quad \exists d \in (0, 1)$$

$$f(1) = f(0) + f'(d) \quad \leftarrow d = \frac{1}{2} \rightarrow \text{GO, } A = 0$$

$$\text{MVT: } \frac{f'(d) - f'(0)}{d - 0} = f''(c) \quad \exists c \in (0, d)$$

$$\frac{f'(d) - f'(0)}{d - 0} = f''(c) \quad \leftarrow d = \frac{1}{2}$$

$$f'(d) = f'(0) + Af''(c)$$

$d < \frac{1}{2}$: same manner

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目录



模式



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